Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 10)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 1 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

0 1 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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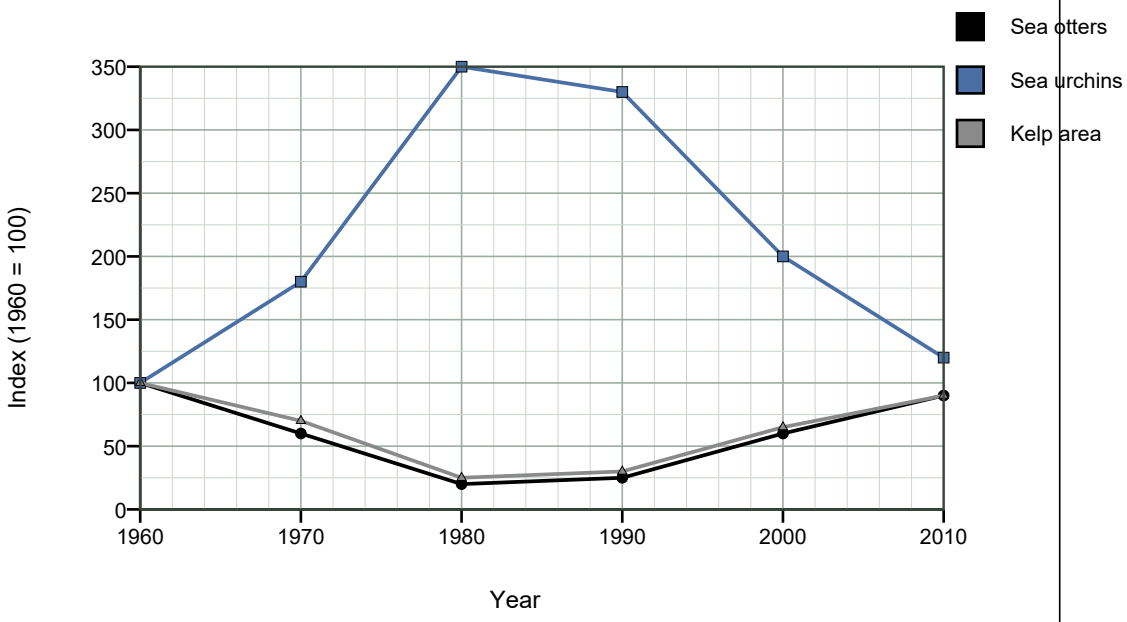
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0 2 Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 2 . 1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

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0 2 . 2 Explain the relationships shown in Figure 1.

[3 marks]

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0 2 . 3 Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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0 2 . 4 Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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0 3 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species .....

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Ecological niche .....

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Community .....

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Biome .....

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0 3 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

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0 4 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 4 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 4 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 4 . 3 Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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0 4 . 4 Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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0 5 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 5 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 5 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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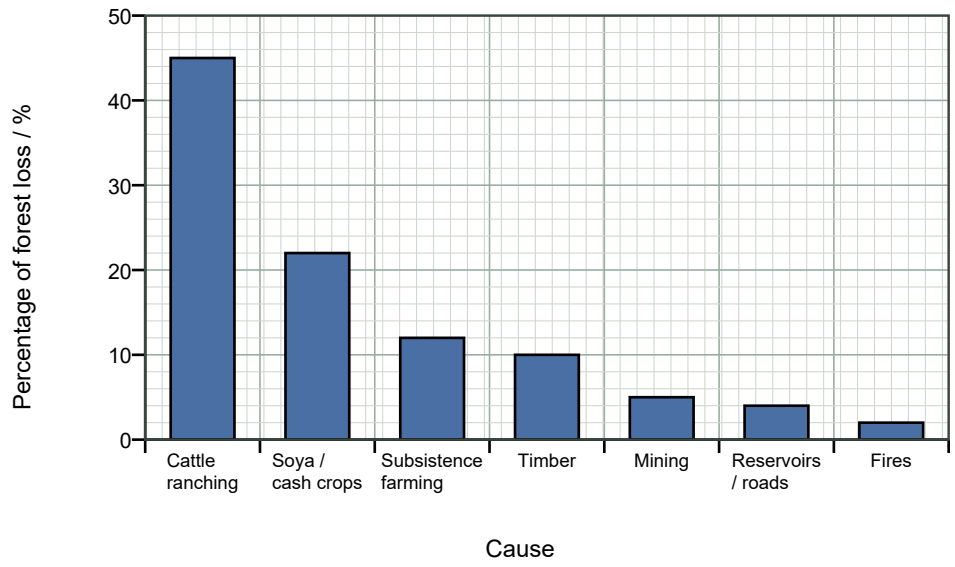
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0 6 Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



0 6 . 1 Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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0 6 . 2 Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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0 6 . 3 Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction .....

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HEP reservoirs .....

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Climate change .....

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0 6 . 4 Describe how a debt-for-nature swap works.

[2 marks]

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0 7 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 7 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 8 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

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0 8 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

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0 8 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

0 8 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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0 9 . 1 Describe the main principles of the circular economy.

[6 marks]

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0 9 . 2 Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

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0 9 . 3 Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

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1 0 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

1 0 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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1 0 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

1 0 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

1 0 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS